

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Easy

Question

In the 1950s and '60s, plant breeders created shorter varieties of wheat and rice plants with improved yields. Kelly Gillespie, Rex Bernardo, and other plant specialists are building on that work by exploring the development of shorter corn varieties. Greater height can allow individual plants to produce more ears of corn. However, greater height also makes the stalks more likely to snap or be uprooted in strong winds before the corn can be harvested. Because of this trade-off, some plant specialists suggest that shorter corn varieties will actually _____

Which choice most logically completes the text?

Answer

- A. help improve yields of harvested corn by being more likely to survive in severe winds.
- B. be more likely to be uprooted due to the weight of the corn on the stalks.
- C. require more land for planting than short varieties of wheat and rice typically do.
- D. begin developing more ears of corn on each plant than the tallest variety of corn currently does.

Correct Answer: A

Rationale

Choice A is the best answer because it presents the conclusion that most logically completes the text’s discussion about a trade-off in growing shorter varieties of corn. The text explains that plant specialists are working to develop shorter corn varieties in an effort to improve yields. The text points out that although greater height in corn plants can lead to more ears of corn per plant, it also makes the stalks more likely to be damaged by strong winds before there is a chance to harvest the corn. This presents a clear trade-off between potential yield (taller plants producing more ears per plant) and harvest reliability (shorter plants surviving until harvest time). Given this information, it is reasonable to infer that shorter corn varieties, while possibly producing fewer ears per plant, would be more likely to survive until harvest time. This survival advantage would result in improved yields of harvested corn because more plants would remain intact, thus offsetting the potential reduction in ears per shorter plant.

Choice B is incorrect. According to the text, greater height makes cornstalks vulnerable to snapping or being uprooted in strong winds, implying that taller plants—not shorter ones—have greater stability problems. The text provides no basis for inferring that shorter corn varieties would be more likely to be uprooted due to corn weight; in fact, the text suggests the opposite relationship between height and stability. Choice C is incorrect because the text doesn’t provide information about the land requirements of either shorter corn varieties or shorter wheat and rice varieties, much less make any comparisons. The text mentions that plant breeders in the 1950s and '60s created shorter varieties of wheat and rice with improved yields and that researchers are now exploring shorter corn varieties, but it doesn’t compare land-use requirements of these crops. Choice D is incorrect because the idea that shorter cornstalks would yield more ears of corn would contradict the relationship between height and corn production described in the text. The text explicitly states that greater height allows individual plants to produce more ears of corn, which suggests that shorter varieties would produce fewer ears per plant, not more. The text therefore provides no basis for concluding that shorter varieties would begin developing more ears than the taller varieties currently do.